1. Introduction to Frequency Response Analysis

Frequency response *is the steady-state response of a system to a sinusoidal input. For a stable LTI system with input* $r(t) = A \sin \omega t$*, the output is* $c_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$*: same frequency, magnitude multiplied by* $|G(j\omega)|$ *and phase shifted by* $\angle G(j\omega)$*.*

$$\boldsymbol{G(j\omega) = G(s)|_{s=j\omega} = M(\omega)\ \angle\varphi(\omega)\ ,\ M = |G(j\omega)|\ ,\ \varphi = \angle G(j\omega)}$$

- ***Advantages:*** *(i) experimental data can be obtained easily; (ii) T.F. can be found from measurements; (iii) stability (relative) is read directly without finding roots; (iv) noise and disturbance effects are visible; (v) design of compensators is simple (graphical); (vi) applicable to systems with transport delay.*
- ***Disadvantages:*** *indirect link with time response; needs a stable open loop for Bode-based conclusions (use Nyquist otherwise).*

2. Frequency Domain Specifications

Specification	Meaning
Resonant peak M_r	Maximum value of $\|M(j\omega)\|$ of closed loop; indicates relative stability (large M_r → large overshoot)
Resonant frequency ω_r	Frequency at which M_r occurs; indicates speed of response
Bandwidth ω_b	Frequency at which $\|M\|$ falls to 0.707 (−3 dB) of its zero-frequency value; large bandwidth → fast response but more noise
Cut-off rate	Slope of log-magnitude beyond ω_b; ability to discriminate noise
Gain crossover ω_{gc}	Frequency where $\|G(j\omega)H(j\omega)\| = 1$ (0 dB)
Phase crossover ω_{pc}	Frequency where $\angle G(j\omega)H(j\omega) = -180°$
Gain margin (GM)	$GM = 1/\|G(j\omega_{pc})H(j\omega_{pc})\|$; in dB: $GM = -20 \log\|GH\|$ at ω_{pc}
Phase margin (PM)	$PM = 180° + \angle G(j\omega_{gc})H(j\omega_{gc})$

Relations for a standard second-order system

$$M_r = 1 / (2\zeta\sqrt{(1-\zeta^2)}) \; (\zeta < 0.707) \; ; \; \omega_r = \omega_n\sqrt{(1-2\zeta^2)}$$

$$\omega_b = \omega_n [\, (1-2\zeta^2) + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}\,]^{1/2}$$

$$PM = \tan^{-1} [\, 2\zeta / \sqrt{(\sqrt{(1+4\zeta^4)} - 2\zeta^2)}\,] \approx 100\zeta \text{ degrees (for } \zeta \leq 0.6)$$

★ Time vs frequency: large M_r ⊠ large M_p; large ω_b ⊠ small t_r; good design: PM = 30°–60°, GM > 6 dB (≥ 2).

3. Bode Diagrams

A Bode plot has two graphs against log ω: (a) magnitude 20 log|G(jω)| in dB and (b) phase angle ∠G(jω) in degrees.

Decade = ratio 10 in frequency; **octave** = ratio 2. 20 dB/decade ≈ 6 dB/octave.

Standard factors

Factor	Magnitude (asymptotic)	Phase
K	$20 \log K$, flat	$0°$ $(K>0)$
$1/(j\omega)^N$	$-20N$ dB/dec, passes 0 dB at $\omega = 1$	$-90°N$
$(j\omega)$	+20 dB/dec through 0 dB at $\omega = 1$	$+90°$
$1/(1 + j\omega T)$	0 dB up to $\omega_c = 1/T$ then -20 dB/dec (actual error -3 dB at ω_c)	$0° \to -45°$ at $\omega_c \to -90°$
$(1 + j\omega T)$	0 dB up to $1/T$, then +20 dB/dec	$0° \to +45° \to +90°$
$1/[1 + 2\zeta(j\omega/\omega_n) + (j\omega/\omega_n)^2]$	0 dB up to ω_n, then -40 dB/dec; resonant peak $-20 \log 2\zeta$ at ω_n for small ζ	$0° \to -90°$ at $\omega_n \to -180°$
$e^{-j\omega T}$ (delay)	0 dB	$-\omega T$ rad (decreasing without bound)

Procedure to draw Bode magnitude plot

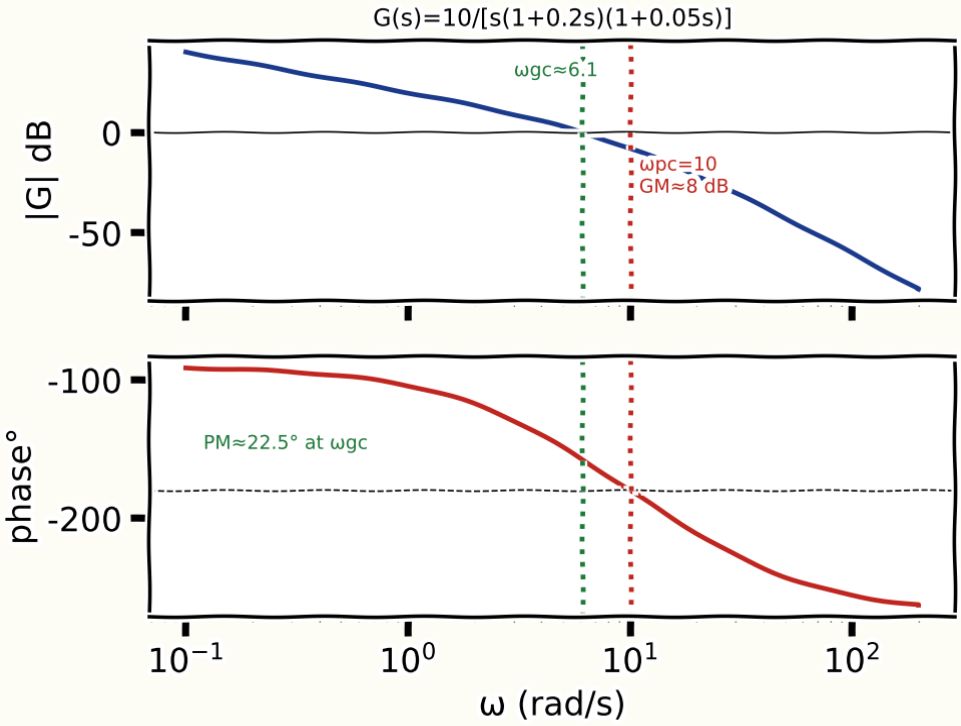
- Write $G(j\omega)$ in standard (time-constant) form; identify K, type N, corner frequencies ($1/T$, ω_n).
- Mark corner frequencies on log scale in increasing order.
- Start: for type 0 draw horizontal line at $20 \log K$; for type N draw line of slope $-20N$ dB/dec passing through the point ($\omega = 1$, $20 \log K$).

- At each corner frequency change the slope: −20 dB/dec for a pole, +20 for a zero (−40 for a quadratic pole pair).
- Phase: evaluate $\varphi(\omega) = \Sigma$ angles at many frequencies (or sketch each factor and add).

Worked example

$G(s) = 10/[\, s(1 + 0.2s)(1 + 0.05s)\,]$. $K = 10$, type 1; corners: $\omega_1 = 1/0.2 = 5$, $\omega_2 = 1/0.05 = 20$ rad/s.

Range of ω	Slope	Remark
$\omega < 5$	−20 dB/dec	At $\omega = 1$ magnitude = 20 log 10 = 20 dB
$5 < \omega < 20$	−40 dB/dec	Gain at 5: 20 − 20 log 5 = 6 dB
$\omega > 20$	−60 dB/dec	Gain at 20 = 6 − 40 log 4 = −18 dB



$$\text{Phase: } \varphi(\omega) = -90^\circ - \tan^{-1}(0.2\omega) - \tan^{-1}(0.05\omega)$$

$$\omega_{pc}: \tan^{-1}(0.2\omega) + \tan^{-1}(0.05\omega) = 90^\circ \Rightarrow 0.01\omega^2 = 1$$

$$\Rightarrow \omega_{pc} = 10 \text{ rad/s}$$

$$|G(j10)| = 10/(10 \cdot \sqrt{5} \cdot \sqrt{1.25}) = 0.4 \Rightarrow GM = 20 \log(1/0.4)$$

$$= 7.96 \text{ dB}$$

$$\omega_{gc} \approx 6.1 \text{ rad/s} \Rightarrow PM = 180^\circ - 90^\circ - 50.6^\circ - 16.9^\circ$$

$$\approx 22.5^\circ$$

*Both GM and PM are positive → **stable** (but PM is low → oscillatory; add lead compensation to improve).*

4. Frequency Domain Specifications and T.F. from the Bode Diagram

Gain margin and phase margin from the Bode plot

- ***PM:*** *read the phase at ω_{gc} (where magnitude curve crosses 0 dB); PM = 180° + $\varphi(\omega_{gc})$.*
- ***GM:*** *read the magnitude at ω_{pc} (phase = −180°); GM = −(magnitude in dB).*
- ***Stability from Bode:*** *GM > 0 dB and PM > 0 ⇒ stable; GM = 0 and PM = 0 ($\omega_{gc} = \omega_{pc}$) ⇒ marginally stable; negative GM/PM ⇒ unstable. For stability, $\omega_{gc} < \omega_{pc}$ (minimum phase systems).*

Determination of transfer function from Bode plot (minimum-phase)

- Initial slope of the magnitude plot gives system type: 0 dB/dec → type 0, −20 → type 1, −40 → type 2.
- Corner frequencies are where slope changes: change of −20 dB/dec → pole $(1 + s/\omega_c)$ in denominator; +20 dB/dec → zero in numerator; −40 → quadratic factor (look at peak to get ζ: $M_r = 1/(2\zeta\sqrt{(1-\zeta^2)})$).
- Find K: type 0 → $20 \log K$ = low-frequency gain; type 1 → the initial −20 dB/dec line (extended) cuts the 0 dB axis at $\omega = K$ (or magnitude at $\omega = 1$ is $20 \log K$); type 2 → cuts at $\omega = \sqrt{K}$.
- Write $G(s) = K \prod(1 + s/z_i) / [\, s^N \prod(1 + s/p_j)\,]$; check the phase curve to confirm minimum phase (and delay if phase falls extra).

Example: initial slope −20 dB/dec, cuts 0 dB at $\omega = 2$ with a further slope change to −40 at $\omega = 10$: $K = 2 \rightarrow G(s) = 2/[s(1 + s/10)] = 20/[s(s + 10)]$.

5. Stability Analysis from Bode Plots

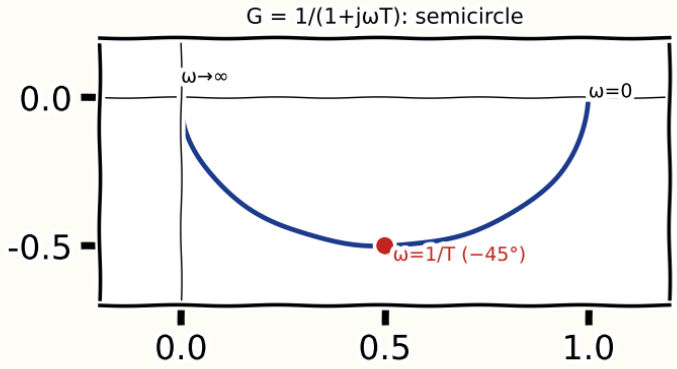
Case	Bode observation	Conclusion
Stable	$\omega_{gc} < \omega_{pc}$, $GM > 0$ dB, $PM > 0$	Closed loop stable
Marginal	$\omega_{gc} = \omega_{pc}$, $GM = 0$, $PM = 0$	Sustained oscillations
Unstable	$\omega_{gc} > \omega_{pc}$, $GM < 0$, $PM < 0$	Growing oscillations

- Increasing K raises the magnitude curve (phase unchanged) → ω_{gc} moves right → PM and GM decrease. Gain for marginal stability: $K_m = K \times 10^{GM/20}$ (e.g., $K = 10$, $GM = 7.96$ dB → $K_{marginal} \approx 25$).
- Phase margin relates to damping: $\zeta \approx PM/100$; GM relates to how much gain increase is tolerable.
- Bode method is not suitable for conditionally stable or open-loop unstable systems → use Nyquist.

★ Merits of Bode plots: easy to draw (straight-line asymptotes), gain & phase on one sheet, multiplication of T.F.s becomes addition, easy compensator design, T.F. from experiments. Demerit: only for minimum-phase systems in the simple form of stability test.

6. Polar Plots

*The **polar plot** of G(jω) is the locus of the tip of the phasor G(jω) (real part vs imaginary part, or magnitude vs phase) as ω varies from 0 to ∞. It gives magnitude and phase on a single plot.*



General shape rules (T.F. with n poles, m zeros, type N)

Type N	Start (ω = 0)	End (ω → ∞)
0	Finite magnitude K on the positive real axis (0°)	Magnitude 0, approaching angle $-90°(n-m)$
1	Magnitude ∞ at angle −90°; asymptote parallel to imaginary axis	Same as above
2	Magnitude ∞ at −180°	$-90°(n-m)$
3	Magnitude ∞ at −270°	$-90°(n-m)$

Procedure

- Write $G(j\omega) = M(\omega)\angle\varphi(\omega)$; find M and φ at $\omega = 0$ and $\omega = \infty$.
- Find the intersection with the real axis (set $Im = 0 \rightarrow \omega_{pc}$, real value $\rightarrow$ GM) and with the imaginary axis ($Re = 0$).
- Plot some intermediate points; join smoothly.

Example: $G(s) = 1/(1 + sT)$: $G(j\omega) = 1/\sqrt{(1+\omega^2T^2)}\ \angle -\tan^{-1}\omega T$; $\omega = 0 \rightarrow 1\angle 0°$; $\omega = 1/T \rightarrow 0.707\angle -45°$; $\omega = \infty \rightarrow 0\angle -90°$. Locus is a **semicircle** in the 4th quadrant of diameter 1.

7. Nyquist Plots and Stability Analysis

The **Nyquist plot** is the polar plot for ω from $-\infty$ to $+\infty$ (the part for negative ω is the mirror image about the real axis). Stability of the closed loop is decided from how it encircles the critical point $\mathbf{-1 + j0}$.

Nyquist contour and mapping

- Nyquist contour: a semicircle of infinite radius enclosing the entire right half s-plane (clockwise direction) with the imaginary axis as its diameter.
- If $G(s)H(s)$ has poles on the $j\omega$ axis (e.g., at origin) indent the contour around them with a small semicircle of radius $\varepsilon \rightarrow 0$ in the RHP. For N poles at origin, the infinite-radius arc in the $G(s)H(s)$ plane covers $N \times 180°$ **clockwise**, from the $\omega = 0^-$ end to $\omega = 0^+$ end.

- *The contour of the 1 + GH function encircling the origin ⇔ the GH plot encircling −1.*

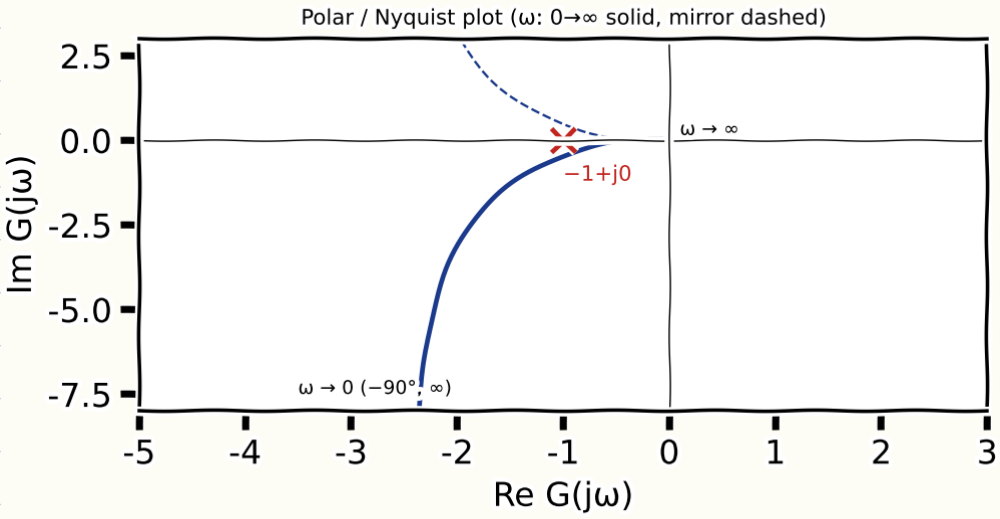
Nyquist stability criterion

$$Z = P - N$$

P = number of open-loop poles in the RHP ; N = number of counter-clockwise encirclements of −1 + j0 by the Nyquist plot ; Z = number of closed-loop poles in the RHP

The closed loop is stable iff Z = 0, i.e., N = P

- *If the open loop is stable (P = 0): closed loop is stable iff the Nyquist plot* ***does not encircle*** *−1 + j0.*
- *If the plot passes through −1 + j0 → marginally stable (poles on jω axis). Clockwise encirclements ⇒ N negative ⇒ Z > 0 unstable.*



For the example G(s) = 10/[s(1 + 0.2s)(1 + 0.05s)], P = 0; the plot crosses the negative real axis at −0.4 (at ω = 10) which is to

the right of -1 → no encirclement → $N = 0 = P$ → ***stable****. Gain margin = 1/0.4 = 2.5 (7.96 dB); a gain increase beyond 2.5 pushes the crossing past -1 → unstable.*

Gain and phase margin from the Nyquist plot

GM = $1 / |G(j\omega_{pc})H(j\omega_{pc})|$ (point where the plot cuts the negative real axis)

PM = $180° + \angle G(j\omega_{gc})H(j\omega_{gc})$ (angle at the point where plot cuts the unit circle)

Feature	Bode plot	Polar plot	Nyquist plot
Axes	Two (dB, degrees vs log ω)	One (Re vs Im)	One (Re vs Im, $-\infty \to \infty$)
Stability	GM, PM from curves	GM, PM	Encirclement of -1 ($Z = P - N$)
Open-loop unstable systems	Not suitable	Not suitable	Suitable
Ease of drawing	Easiest	Moderate	Needs care at origin/infinite arcs

Important Questions

- Define M_r, ω_r, ω_b, GM, PM; derive M_r and ω_r for a 2nd-order system. (10M)

- *Draw the Bode plot for the given G(s)H(s); find GM, PM and comment on stability. (10M)*
- *Determine the T.F. from the given Bode magnitude plot. (5M/10M)*
- *Sketch the polar plot for the given T.F. (5M)*
- *State Nyquist criterion; draw the Nyquist plot and find the range of K for stability. (10M)*